

Exponent Laws with Variable Bases

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	3
Lesson	Thread A · Day 1 — Exponent Laws with Variable Bases	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.AR.1.1

2. Standards

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Must. Whole-number exponents on monomial bases: multiply and divide monomials, and raise a monomial to a power, generating an equivalent expression each time.

Must not. Whole-number exponents today; the negative ones are tomorrow. No rational exponents, and monomials only — nothing with a plus sign in it.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — $x^2 \cdot x^3$ is expanded into five letters before the rule is restated, so the law is recognized as the one from 3.01 rather than met as a new one.
MA.K12.MTR.3.1	Complete tasks with mathematical fluency.	Boards 1, 2 and 7 — one move each, written and up on the cue; the law has to be automatic before the negative exponents arrive tomorrow.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 5 — when the room splits over whether the 5 gets squared, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can multiply and divide monomials, and raise a monomial to a power, by applying the laws of exponents to variable bases.

Essential question. Do the laws of exponents change when the base is a variable?

Building on. MA.7.NSO.1.1 and MA.8.NSO.1.3 (3.01–3.07) — the laws with numerical bases; Grade 7 — $x \cdot x \cdot x = x^3$ and $3x$ means $3 \cdot x$.

Working toward. T-A2 — negative exponents on variable bases and several laws at once; MA.912.NSO.1.2 — rational exponents on algebraic expressions in Algebra 1.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	5	Two minutes silent, then reveal. One sentence of reteach per question at most.	Four retrieval questions, alone, no notes.
Notes	11	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	13	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	53		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Rachel writes fifteen factors and Justina writes $2,744a^9b^3$. Which form is easier to check, and which is easier to use in the next step?	DOK 2
$(2x^3)^3$ has a 2 inside the parentheses. Why does the exponent reach the 2 instead of stopping at the x?	DOK 3
$x^3 \cdot x^n = x^{11}$ has exactly one answer. Would $x^3 \cdot y^n = x^3y^8$ also have exactly one? Say what changed.	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$x^5 \cdot x^8$	x^{13}	x^{40} — multiplied the exponents [B1G-M 8.AR.1.1 misconception: misapplied laws]; $2x^{13}$ — an invented coefficient
2	$(y^5)^3$	y^{15}	y^8 — added the exponents [8.AR.1.1: power of a power confused with a product]
3	A cube has edges $2x^3$ centimeters long. Its surface area is $24x^6$ square centimeters. What is the volume of the cube, in cubic centimeters?	$8x^9$	$2x^9$ — the coefficient was not raised [B1G-M 8.AR.1.1: power of a product]; $8x^6$ — exponents added instead of multiplied [confused with product of powers]; anything built from $24x^6$ — the unneeded figure was used
4	$(3a^2b)(4ab^5)$	$12a^3b^6$	$7a^3b^6$ — coefficients added instead of multiplied [8.AR.1.1: coefficients are factors]; $12a^2b^5$ — the factors a and b with exponent 1 were dropped
5	$(5m^3)^2$ Which expression is equivalent?	$A - 25m^6$	(B) $5 \cdot 2$ instead of 5^2 — the exponent was multiplied into the coefficient [B1G-M 8.AR.1.1 misconception: multiplying the exponent into the base]; (C) $3 + 2$ instead of $3 \cdot 2$ [power of a power confused with a product of powers]; (D) the coefficient was not raised at all [8.AR.1.1: power of a product]
6	$\frac{12x^8}{3x^2}$ Which expression is equivalent?	$B - 4x^6$	(A) $8 + 2$ — added the exponents [quotient law confused with product law]; (C) $12 - 3$ — the coefficients were subtracted instead of divided [8.AR.1.1: coefficients are factors]; (D) $8 \div 2$ — the exponents were divided instead of subtracted [B1G-M 8.AR.1.1: misapplied quotient law]
7	$\frac{c^{10}d^3}{c^2d^3}$	c^8	c^8d — d^0 read as d [8.AR.1.1: zero exponent]; 0 or $c^8 \cdot 0$ — d^0 read as 0 [8.AR.1.1: zero exponent, $d^0 = 1$]
8	Find the value of n : $x^3 \cdot x^n = x^{11}$	$n = 8$	14 — added 3 to 11; 33 — multiplied [8.AR.1.1: product law is additive in the exponents]
9	Rachel writes $(14a^3b)^3$ as $14 \cdot 14 \cdot 14 \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot b \cdot b \cdot b$. Justina writes it as $2,744a^9b^3$. Who is correct? Explain.	Both are correct; $14^3 = 2,744$, nine a's make a^9, and $b \cdot b \cdot b = b^3$.	'Justina only' — the expanded form is not recognized as equivalent [8.AR.1.1: generate equivalent expressions]; 'Rachel only' or ' $14a^9b^3$ ' — the coefficient was not cubed [B1G-M 8.AR.1.1: power of a product]; ' $42a^9b^3$ ' — the coefficient multiplied by 3 instead of cubed

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Let the student write the expansion under the first two boards — $x \cdot x \cdot x \cdot x \cdot x$ for x^5 — and count. Boards 3 and 4 are done in two passes: circle the coefficients and handle them, then the letters.

ELL. Monomial and coefficient are new nouns. Write $12a^3b^6$ once with COEFFICIENT under the 12 and BASES under the a and the b, and leave it up through the round.

Enrichment. Ask for $(2x^2y)^3 \cdot (3xy^4)^2$ as a single monomial, and for the n that makes $(x^n)^4 = x^{20}$.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to say that both forms are the same expression and show that 14^3 is 2,744. Read the boards, not the papers.

Vocabulary. **monomial**; **coefficient**; **like bases**

Materials. Whiteboards and markers. Calculators are allowed on everything; there is little for them to do today beyond 14^3 on whiteboard 9.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Multiply powers: variable bases — TR9; Powers of a power: variable bases — 2CU; Divide monomials with positive exponents — C8L.